

Revision Game Cards — 1.1 Processors, Input, Output & Storage

Print and cut along the borders. ✂ Loop cards: each player reads the bottom "Who has...?"; whoever holds the matching "I have..." reads it out, then their own clue — the chain visits every card and loops back to the start.

Loop cards (dominoes) — cut out & deal

I HAVE 1 / 16 Register WHO HAS... the small, fast memory on or near the CPU that holds frequently used data to cut slow RAM fetches?	I HAVE 2 / 16 Cache WHO HAS... the technique of overlapping the fetch, decode and execute of consecutive instructions to raise throughput?
I HAVE 3 / 16 Pipelining WHO HAS... an independent processing unit able to run its own fetch-decode-execute cycle?	I HAVE 4 / 16 Core WHO HAS... the many-core processor that performs the same operation on large data sets in parallel?
I HAVE 5 / 16 GPU WHO HAS... carrying out multiple computations simultaneously by splitting a task across cores?	I HAVE 6 / 16 Parallel processing WHO HAS... the architecture with one shared memory and bus for both data and instructions?

I HAVE

Von Neumann

WHO HAS...

the architecture with separate memories and buses for data and instructions, allowing simultaneous access?

I HAVE

Harvard

WHO HAS...

the processor type with many complex, variable-length instructions and complexity in hardware (e.g. x86)?

I HAVE

CISC

WHO HAS...

the processor type with few simple fixed-length instructions aiming for one per cycle, used in ARM mobile devices?

I HAVE

RISC

WHO HAS...

the CPU component that performs arithmetic, logic and shift operations?

I HAVE

ALU

WHO HAS...

the CPU component that decodes instructions, generates control signals and synchronises with the clock?

I HAVE

Control Unit

WHO HAS...

the register holding the address of the next instruction, incremented each cycle?

I HAVE

Program Counter (PC)

WHO HAS...

the register that stores the immediate result of an ALU calculation?

I HAVE

Accumulator (ACC)

WHO HAS...

the use of secondary storage as extra RAM when physical RAM is full?

I HAVE

Virtual memory

WHO HAS...

the property of memory that loses its contents when power is removed (e.g. RAM)?

I HAVE

Volatile

WHO HAS...

the small, very fast store inside the CPU that holds data during processing?

Taboo cards — describe the term without the banned words

MAR (Memory Address Register)

Don't say:

- address
- location
- memory
- read

Cache

Don't say:

- fast
- CPU
- frequently
- RAM

Pipelining

Don't say:

- overlap
- fetch
- execute
- instructions

Von Neumann

Don't say:

- shared
- memory
- bus
- instructions

GPU

Don't say:

- parallel
- cores
- graphics
- pixels

Virtual memory

Don't say:

- RAM
- secondary
- disk
- pages

Volatile

Don't say:

- power
- lost
- RAM
- contents

Optical storage

Don't say:

- laser
- CD
- DVD
- pits

Bingo — word bank & ready-to-cut cards

Word bank (students fill a blank 3×3 grid, or use the pre-filled cards below): ALU, Control Unit, Program Counter, MAR, MDR, CIR, Accumulator, Cache, Pipelining, Core, GPU, Clock speed, Von Neumann, Harvard, CISC, RISC, Magnetic storage, Flash/SSD, Optical storage, Virtual memory

BINGO 1

Harvard	MAR	Von Neumann
Clock speed	Program Counter	Cache
RISC	Pipelining	Accumulator

BINGO 2

Control Unit	Accumulator	Program Counter
Von Neumann	CIR	Optical storage
Clock speed	RISC	Core

BINGO 3

Magnetic storage	Harvard	Cache
MDR	Virtual memory	RISC
Flash/SSD	Program Counter	GPU

BINGO 4

RISC	Flash/SSD	Clock speed
Cache	CISC	Magnetic storage
Pipelining	Virtual memory	Accumulator

BINGO 5

Harvard	MAR	Flash/SSD
Magnetic storage	Accumulator	Von Neumann
Virtual memory	CISC	Optical storage

BINGO 6

RISC	Von Neumann	Flash/SSD
Harvard	Core	CIR
Pipelining	GPU	Program Counter

Bingo — caller's list (teacher: read the clue, students cross off the term)

#	Clue to read aloud	Answer
1	Performs arithmetic, logic and shift operations.	ALU
2	Decodes instructions and generates control signals.	Control Unit
3	Holds the address of the next instruction.	Program Counter
4	Holds the address of the location to read from or write to.	MAR
5	Holds the data or instruction just fetched from memory.	MDR
6	Holds the current instruction while it is decoded and executed.	CIR
7	Stores the immediate result of an ALU calculation.	Accumulator
8	Small fast memory on/near the CPU holding frequently used data.	Cache
9	Overlapping fetch, decode and execute to raise throughput.	Pipelining
10	An independent unit that runs its own FDE cycle.	Core
11	Many-core processor applying one operation across large data sets.	GPU
12	Pulses per second that synchronise the CPU, measured in Hz.	Clock speed
13	One shared memory and bus for data and instructions.	Von Neumann
14	Separate memories and buses for data and instructions.	Harvard
15	Many complex, variable-length instructions; complexity in hardware.	CISC
16	Few simple fixed-length instructions; used in ARM devices.	RISC
17	Stores data as magnetised patterns on a spinning platter.	Magnetic storage
18	Non-volatile EEPROM storage with no moving parts.	Flash/SSD
19	A laser reads pits and lands via reflected light.	Optical storage
20	Uses secondary storage as extra RAM when RAM is full.	Virtual memory

Quick-fire — clue & answer (self-test / quiz-quiz-trade)

#	Clue	Answer
1	The bus that is unidirectional, carrying addresses from CPU to memory.	Address bus
2	The register pair you must never swap: address vs data.	MAR and MDR
3	The step in fetch most often forgotten, along with copying MDR to CIR.	Incrementing the PC
4	Non-volatile memory holding the bootstrap/BIOS firmware.	ROM
5	Volatile memory holding currently running programs and data.	RAM
6	The cache level that is fastest and smallest, on the core.	L1 cache
7	Excessive paging between RAM and disk that grinds a system to a halt.	Disk thrashing
8	The part of a task that must run sequentially, limiting the speed-up from adding cores.	Serial (sequential) fraction
9	Storage access type of tape: you must read through in sequence.	Serial (sequential) access
10	Technique flash memory uses to spread writes and prolong cell life.	Wear levelling
11	Output device that builds objects layer by layer.	3D printer
12	The general rule for 'choose a device' questions: link each characteristic to the requirement.	Scenario / requirement (and compare)